

ERGOGUARD CHAIR

(CONTEXT AWARE POSTURE + STRESS + FATIGUE PREVENTION SYSTEM)

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PROBLEM STATEMENT

- 70% of IT workers suffer from chronic back/neck pain
- Average sitting time: ~9 hours/day
- Existing solutions fail:
 - Wearables (e.g., Apple Watch): can't measure posture/weight distribution
 - Camera-based monitoring: privacy concerns
 - Fixed reminders: not context-aware
- Impact: productivity loss + rising healthcare costs



PROPOSED SOLUTION

- ErgoGuard Chair is a hardware + software product that:
- Detects posture in real-time using seat + backrest pressure sensors
- Tracks sedentary behavior using micro-movement sensing (IMU)
- Gives adaptive feedback via vibration + LED status
- Provides analytics through a web dashboard



WHY ERGOGUARD IS DIFFERENT

- Wearables measure vitals; ErgoGuard measures posture + load distribution
- No camera required ? privacy-preserving
- Closed-loop intervention (LED/vibration), not just reminders
- Auto-calibration baseline per user (no manual tuning)
- Designed as a scalable product for offices/call centers/students



SYSTEM ARCHITECTURE (PROPOSED)

Sensors:

ESP32 (edge processing + WiFi)

Hardware:

- 8x FSR pressure sensors (seat + backrest)
- MPU6050 IMU (tilt + micro-movement)
- Optional: MQ135 + DHT22 for environment
- Actuators: vibration motors + WS2812B LED strip

Software:

- Firmware (ESP32) + Backend (Node.js/MongoDB) + Frontend (React)



HARDWARE OVERVIEW

Core components:

- ESP32 (WiFi + processing)
- 8x FSR sensors (pressure mapping)
- MPU6050 (movement + tilt)
- Vibration motors (haptic feedback)
- RGB LED strip (status feedback)
- Optional: MQ135 + DHT22 (air/temperature)



CORE ALGORITHM (ERGO-STRESS SCORE)

Base score = 100

Penalties from sensor fusion:

- Posture deviation (forward lean / side lean / slouch)
- Asymmetry (left-right / front-back weight imbalance)
- Sedentary risk (low micro-movement + long sitting duration)
- Optional environment penalty (poor air quality / temperature)

Final Score = 100 - (weighted penalties)



KEY FEATURES (PLANNED)

Monitoring:

- Posture score, asymmetry score, sedentary score, combined Ergo-Stress score

Interventions:

- LED: Green (good), Amber (attention), Red (critical)
- Vibration: intensity/pattern proportional to risk

Dashboard:

- Real-time view + history trends + posture breakdown



INNOVATION, IMPACT & CONCLUSION

- Multi-modal sensor fusion : unified ergonomic risk score
- Adaptive intervention policy (not fixed reminders)
- Privacy-preserving design (no camera)
- Auto-calibration for different body types

Impact:

- Better posture awareness, reduced fatigue, better break habits
- Scalable for IT offices/call centers/students



THANKYOU!

